Name: _____ / 25

30 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 25 points possible.

1. Consider the function $f(x,y) = x^2 + xy + y^2 - 3x$.

   (a) (4 points) Find all critical points of f.

   Solution

   $$f_x = 2x + y - 3 = 0, \qquad f_y = x + 2y = 0.$$

   The second equation gives $x = -2y$; substituting into the first, $-4y + y - 3 = 0$, so $y = -1$ and $x = 2$.

   $$\boxed{(2, -1)}$$

   (b) (3 points) Classify the critical points using $D(x_0,y_0) = f_{xx}(x_0,y_0)f_{yy}(x_0,y_0) - (f_{xy}(x_0,y_0))^2$.

   Solution

   $f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 1$, so

   $$D(2,-1) = (2)(2) - 1^2 = 3 > 0 \quad \text{and} \quad f_{xx}(2,-1) = 2 > 0.$$

   $$\boxed{f(2,-1) \text{ is a local minimum}}$$

2. (6 points) Find an equation for the tangent plane to the surface $-4x^3+12y^2+4z=-4$ at the point $(2,1,4)$.

Solution

Solving for z: $4z=4x^3-12y^2-4$, so $z=x^3-3y^2-1$.

$$\frac{\partial z}{\partial x}=3x^2, \quad \left.\frac{\partial z}{\partial x}\right|_{(2,1)}=12; \qquad \frac{\partial z}{\partial y}=-6y, \quad \left.\frac{\partial z}{\partial y}\right|_{(2,1)}=-6.$$

$$\boxed{z=4+12(x-2)-6(y-1)}$$

3. Consider the function $f(x,y)=e^x\cos(y)$.

(a) (6 points) Find the directional derivative of f in the direction of the vector $\langle 3,-4\rangle$ at the point $(0,\frac{\pi}{2})$.

Solution

$$\nabla f=\langle e^x\cos y,-e^x\sin y\rangle, \qquad \nabla f\left(0,\tfrac{\pi}{2}\right)=\langle 0,-1\rangle.$$

$|\langle 3,-4\rangle|=5$, so $\mathbf{u}=\left\langle \frac{3}{5},-\frac{4}{5}\right\rangle$ and

$$D_{\mathbf{u}}f\left(0,\tfrac{\pi}{2}\right)=\langle 0,-1\rangle\cdot\left\langle \tfrac{3}{5},-\tfrac{4}{5}\right\rangle=0+\tfrac{4}{5}=\boxed{\frac{4}{5}}$$

(b) (3 points) Find the direction of the maximum rate of change of f at the point $(0,\frac{\pi}{2})$.

Solution

The maximum rate of change is in the direction of the gradient:

$$\boxed{\nabla f\left(0,\tfrac{\pi}{2}\right)=\langle 0,-1\rangle}$$

(c) (3 points) Find the maximum rate of change of f at the point $(0,\frac{\pi}{2})$.

Solution

$$\left|\nabla f\left(0,\tfrac{\pi}{2}\right)\right|=\sqrt{0^2+(-1)^2}=\boxed{1}$$